# Tensor calculus (introduction)

! here only Cartesian coordinate systems

Notation: Scalars: $a, b$

Vectors: $\underset{\sim}{a}, \underset{\sim}{b}$ (Tensors 1st order)

Tensors 2nd/3rd order: $\underset{\sim}{\underline{A}}, \underset{\sim}{\underline{B}}$ or $\underline{\underline{A}}, \underline{\underline{B}}$

Tensors 4th order $\underset{\sim}{\underline{\underline{\mathbb{A}}}}$ or $\underline{\underline{\mathbb{A}}}$

Matrices: $\underline{\underline{A}}$

## Summation convention

"Double indices in a product are summed up"

$$\underset{\sim}{v} = v_1 \underset{\sim}{e}_1 + v_2 \underset{\sim}{e}_2 + v_3 \underset{\sim}{e}_3 = \sum_{i=1}^{3} v_i \underset{\sim}{e}_i$$

$$= v_i \underset{\sim}{e}_i$$

## Rules for indices:

1) Every index appears maximally twice in product term

Expl:

- $a_i b_i c_i$ → not defined
- $a_i b_i c_j = a_1 b_1 c_j + a_2 b_2 c_j + a_3 b_3 c_j$
- $a_i b_i + c_i d_i = a_1 b_1 + c_1 d_1 + a_2 b_2 + c_2 d_2 + \ldots$

2) Free index $\hat{=}$ index, which appears only once in a product term

! number and letters identical in each additive term

Expl.: • $a_i b_i c_j + d_i e_i f_k \longrightarrow$ not defined

• $a_i b_i c_j + d_i e_i f_j = a_1 b_1 c_j + d_1 e_1 f_j + a_2 b_2 c_j + \dots$

3) Muted index $\hat{=}$ summation indices can be renamed arbitrarily

Expl.: • $V_i W_i = V_k W_k$

• $a_i b_i c_j + d_i e_i f_j = a_i b_i c_j + d_k e_k f_j$

## Different notations

$\underset{\sim}{a} \hat{=}$ Symbolic not.

$a_i \underset{\sim}{e}_i \hat{=}$ component not./complete not. } index notation

$a_i \hat{=}$ coefficient not. } index notation

$\begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} \hat{=}$ vector notation

$\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \hat{=}$ matrix notation

$\longrightarrow [A_{11} \; A_{22} \; A_{12} \; A_{21}]^T \hat{=}$ reduced matrix not.

## Rules

**Vectors:** $\underset{\sim}{a} = a_i \underset{\sim}{e}_i \hat{=} a_i$

Here: Cartesian coordinates ⇝ calculation in coefficient notation

**Norm:** $|\underset{\sim}{a}| = \sqrt{a_1^2 + a_2^2 + a_3^2}$

**Sum:** $\underset{\sim}{a} + \underset{\sim}{b} = a_1 \underset{\sim}{e}_1 + a_2 \underset{\sim}{e}_2 + a_3 \underset{\sim}{e}_3 + \ldots$

$$\ldots + b_1 \underset{\sim}{e}_1 + b_2 \underset{\sim}{e}_2 + b_3 \underset{\sim}{e}_3$$

$$= (a_1 + b_1) \underset{\sim}{e}_1 + (a_2 + b_2) \underset{\sim}{e}_2 + (a_3 + b_3) \underset{\sim}{e}_3$$

$$= (a_i + b_i) \underset{\sim}{e}_i$$

**Scalar product:** (reduces order)

$$\underset{\sim}{a} \cdot \underset{\sim}{b} = |\underset{\sim}{a}||\underset{\sim}{b}| \cos \sphericalangle(\underset{\sim}{a}, \underset{\sim}{b})$$

$$= (a_i \underset{\sim}{e}_i) \cdot (b_j \underset{\sim}{e}_j)$$

$$= a_i b_j \delta_{ij} = a_i b_j \delta_{ji} = a_i b_i$$

Kronecker-Delta

$$\delta_{ij} = \begin{cases} 1, & i = j \\ 0, & i \neq j \end{cases}$$

$$\rightsquigarrow b_j \delta_{ji} = b_1 \delta_{1i} + b_2 \delta_{2i} + b_3 \delta_{3i}$$

$i = 1 \rightsquigarrow b_j \delta_{ji} = b_1 \underbrace{\delta_{11}}_{=1} + b_2 \underbrace{\delta_{21}}_{=0} + b_3 \underbrace{\delta_{31}}_{=0} = b_1$

$i = 2 \rightsquigarrow \quad = b_2$

$$\hookrightarrow b_j \delta_{ji} = b_i$$

Vector product: $\underset{\sim}{a} \times \underset{\sim}{b} = (a_2 b_3 - a_3 b_2)\,\underset{\sim}{e}_1$

(keeps order)

$$+ (a_3 b_1 - a_1 b_3)\,\underset{\sim}{e}_2$$

$$+ (a_1 b_2 - a_2 b_1)\,\underset{\sim}{e}_3$$

Dyadic product: $\underset{\sim}{a} \otimes \underset{\sim}{b} = (a_i \underset{\sim}{e}_i) \otimes (b_k \underset{\sim}{e}_k)$

(increases order)

$$= a_i b_k \underset{\sim}{e}_i \otimes \underset{\sim}{e}_k$$

$$\hat{=}\ a_i \cdot b_k$$

coefficient matrix of $\underset{\sim}{a} \otimes \underset{\sim}{b}$:

$$\begin{bmatrix} a_1 b_1 & a_1 b_2 & a_1 b_3 \\ a_2 b_1 & a_2 b_2 & a_2 b_3 \\ a_3 b_1 & a_3 b_2 & a_3 b_3 \end{bmatrix}$$

# Tensors

$\underline{A} = A_{ij\ldots n}\, \underset{\sim}{e}_i \otimes \underset{\sim}{e}_j \otimes \ldots \otimes \underset{\sim}{e}_n$ (Number of indices, i.e. base vectors defines order)

Expl.

2nd order: $\underline{A} = A_{ij}\, \underset{\sim}{e}_i \otimes \underset{\sim}{e}_j$

4th order: $\mathbb{A} = \mathbb{A}_{ijkl}\, \underset{\sim}{e}_i \otimes \underset{\sim}{e}_j \otimes \underset{\sim}{e}_k \otimes \underset{\sim}{e}_l$

Sum: (component-wise)

$\underline{A} + \underline{B} = (A_{ij} + B_{ij})\, \underset{\sim}{e}_i \otimes \underset{\sim}{e}_j$

Single contraction / inner product

$$\underline{A}\,\underline{B} = \left(A_{ij}\, \underset{\sim}{e}_i \otimes \underset{\sim}{e}_j\right)\left(B_{kl}\, \underset{\sim}{e}_k \otimes \underset{\sim}{e}_l\right)$$

($\underset{\sim}{e}_j \otimes \underset{\sim}{e}_k$: inner product)

$$= A_{ij}\, B_{kl}\, (\underset{\sim}{e}_j \cdot \underset{\sim}{e}_k)\, \underset{\sim}{e}_i \otimes \underset{\sim}{e}_l$$

$$= A_{ij}\, B_{kl}\, \delta_{jk}\, \underset{\sim}{e}_i \otimes \underset{\sim}{e}_l = A_{ij}\, \underbrace{\delta_{jk}\, B_{kl}}_{B_{jl}}\, \underset{\sim}{e}_i \otimes \underset{\sim}{e}_l$$

$$= A_{ij}\, B_{jl}\, \underset{\sim}{e}_i \otimes \underset{\sim}{e}_l$$

$$\hat{=}\, A_{ij}\, B_{jl}$$

Double contraction:

- $\underline{\underline{\mathbb{C}}} : \underline{\underline{\mathbb{B}}} = \mathbb{C}_{ijkl}\, \underset{\sim}{e}_i \otimes \underset{\sim}{e}_j \otimes \underset{\sim}{e}_k \otimes \underset{\sim}{e}_l \; \mathbb{B}_{op}\, \underset{\sim}{e}_o \otimes \underset{\sim}{e}_p$

$$= \mathbb{C}_{ijkl}\, \mathbb{B}_{op}\, \delta_{ko}\, \delta_{lp}\, \underset{\sim}{e}_i \otimes \underset{\sim}{e}_j$$

$$= \mathbb{C}_{ijkp}\, \mathbb{B}_{kp}\, \underset{\sim}{e}_i \otimes \underset{\sim}{e}_j$$

$$\hat{=}\; \mathbb{C}_{ijkp}\, \mathbb{B}_{kp}$$

Scalar product (special case: complete contraction)

$$\underline{\underline{\mathbb{A}}} \cdot \underline{\underline{\mathbb{B}}} = \left( \mathbb{A}_{ij}\, \underset{\sim}{e}_i \otimes \underset{\sim}{e}_j \right) \cdot \left( \mathbb{B}_{kl}\, \underset{\sim}{e}_k \otimes \underset{\sim}{e}_l \right)$$

$$= \mathbb{A}_{ij}\, \mathbb{B}_{kl} \left( \underset{\sim}{e}_i \cdot \underset{\sim}{e}_k \right) \left( \underset{\sim}{e}_j \cdot \underset{\sim}{e}_l \right)$$

$$= \mathbb{A}_{ij}\, \mathbb{B}_{kl}\, \delta_{ik}\, \delta_{jl}$$

$$= \mathbb{A}_{il}\, \mathbb{B}_{il}$$

$$= \underline{\underline{\mathbb{A}}} : \underline{\underline{\mathbb{B}}}$$

Dyadenprodukt:

$$\underline{\underline{\mathbb{A}}} \otimes \underline{\underline{\mathbb{B}}} \hat{=} \mathbb{A}_{ij}\, \mathbb{B}_{kl}$$

$$\underline{\underline{\underline{\underline{\mathbb{A}}}}} \otimes \underline{\underline{\underline{\underline{\mathbb{B}}}}} \hat{=} \mathbb{A}_{ijkl}\, \mathbb{B}_{mnop}$$

## Special tensors

**Zero-Tensor:** $\underline{\underline{O}} \quad \left(\rightsquigarrow \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}\right)$

**Identity tensor:** $\underline{\underline{A}}\,\underline{\underline{I}} = \underline{\underline{A}}$

($\underline{\underline{I}}$ or $\underline{\underline{1}}$)

- 2nd order: $\underline{\underline{I}} = \delta_{ij}\, \underset{\sim}{e}_i \otimes \underset{\sim}{e}_j \mathrel{\hat{=}} I_{ij}$
- 4th order: $\underline{\underline{\underline{\underline{I}}}} \mathrel{\hat{=}} \mathbb{I}_{mnkl} = \delta_{mk}\,\delta_{nl}$

**Transposed tensor:** (interchange indices)

Expl:

$$\underline{\underline{A}} \mathrel{\hat{=}} A_{ij} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \rightarrow \underline{\underline{A}}^T \mathrel{\hat{=}} A_{ji} = \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix}$$

**Inverse tensor:** Def.: $\underline{\underline{T}}\,\underset{\sim}{v} = \underset{\sim}{w} \Leftrightarrow \underset{\sim}{v} = \underline{\underline{T}}^{-1}\,\underset{\sim}{w}$ exists

$$\rightsquigarrow \quad \underline{\underline{T}}^{-1}\,\underline{\underline{T}} = \underline{\underline{I}}$$

## special operators

**Trace:** $\operatorname{tr}[\underline{A}] = \underline{I} \cdot \underline{A} = A_{ii}$

(reduces order)

**Gradient:** $\operatorname{grad}[\underset{\sim}{u}] = \dfrac{\partial \underset{\sim}{u}}{\partial \underset{\sim}{x}} = \dfrac{\partial \underset{\sim}{u}}{\partial x_j} \otimes \underset{\sim}{e}_j = \dfrac{\partial u_i}{\partial x_j} \underset{\sim}{e}_i \otimes \underset{\sim}{e}_j$

(increases order)

$$\rightsquigarrow \{\operatorname{grad} \underset{\sim}{u}\}_{ij} = \frac{\partial u_i}{\partial x_j} =: u_{i,j}$$

**Divergence:** $\operatorname{div}[\underset{\sim}{u}] \hat{=} \dfrac{\partial u_1}{\partial x_1} + \dfrac{\partial u_2}{\partial x_2} + \dfrac{\partial u_3}{\partial x_3} = u_{i,i}$

(reduces order)

$$\operatorname{div}[\underline{A}] \hat{=} \frac{\partial A_{i1}}{\partial x_1} + \frac{\partial A_{i2}}{\partial x_2} + \frac{\partial A_{i3}}{\partial x_3}$$

$$= A_{ij,j}$$